taking photographs and also taking soil/rock samples for later analysis.

i Describe how sensors, actuators and a microprocessor can be used to take samples from the planet's surface.

ii Describe three uses of the cameras on this autonomous robot.

c Describe the advantages and disadvantages of using autonomous robots in both undersea and outer space exploration.

d Give two other examples of where autonomous robots could be used.

6.3 Artificial intelligence (AI)

6.3.1 Introduction

Artificial intelligence (AI) is a branch of computer science dealing with the simulation of intelligent human behaviour by a computer. This is often referred to as the **cognitive** functions of the human brain (that is, the mental process of acquiring knowledge and understanding through thought, experience and the five senses). All of these cognitive functions can be replicated in a machine, and they can be measured against human benchmarks such as reasoning, speech and sight.

6.3.2 Characteristics of AI

Essentially, AI is really just a collection of rules and data, and the ability to reason, learn and adapt to external stimuli. AI can be split into three categories:

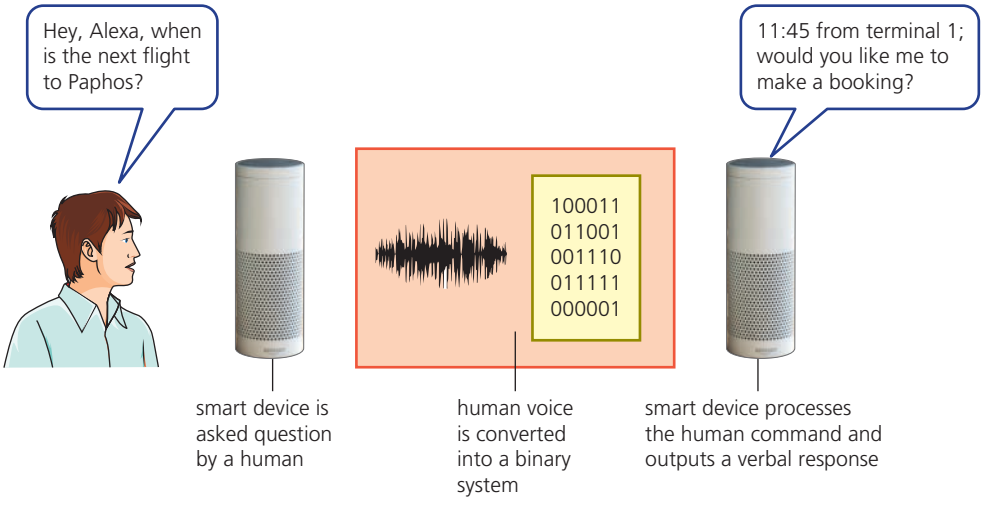
- narrow AI – this occurs when a machine has superior performance to a human when doing one specific task
- general AI – this occurs when a machine is similar (not superior) in its performance to a human doing a specific task
- strong AI – this occurs when a machine has superior performance to a human in many tasks.

Reasoning is the ability to draw reasoned conclusions based on given data/situations. Deductive reasoning is where a number of correct facts are built up to form a set of rules which can then be applied to other problems (for example, if AI is used to produce the perfect cup of tea based on a number of facts, the machine will learn from the experience and apply its new rules in the making of a cup of coffee, hot chocolate, and so on – modifying its methodology where necessary). By carrying out a sequence of steps, the AI machine can learn, and next time it will know how to do the task more effectively and even apply it to a novel/new situation. Thus the AI system is capable of learning and adapting to its surroundings. AI can very quickly discern patterns (which in some cases, humans cannot) and then make predictions by adapting to the new data. How all this is done is beyond the scope of this book (interested readers can find out more about this topic by consulting *Cambridge International AS and A Level Computer Science,* ISBN: 9781510457584).

Examples of AI include:

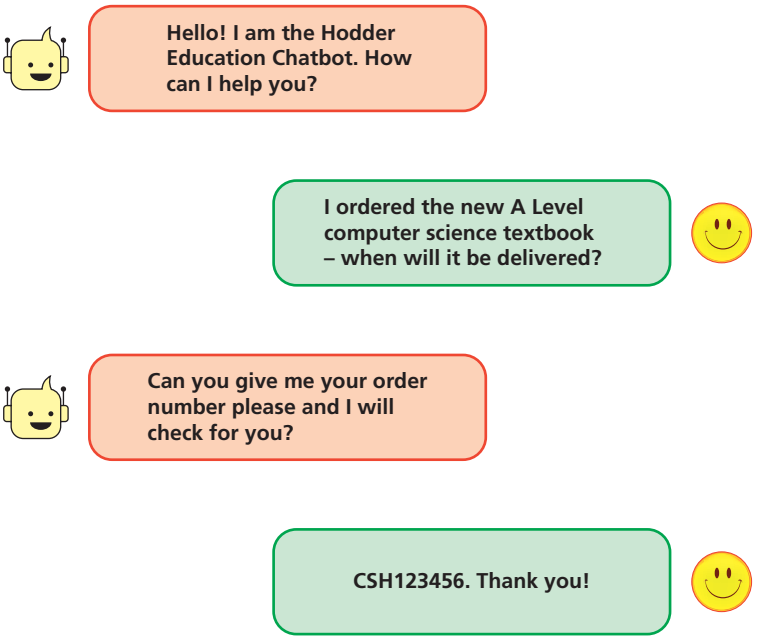
- news generation based on live news feeds
- smart home devices (such as Amazon Alexa, Google Now, Apple Siri and Microsoft Cortana):
  - the AI device interacts with a human by recognising verbal commands
  - it learns from its environment and the data it receives

– the device becomes increasingly sophisticated in its responses, thus showing the ability to use automated repetitive learning



▲ **Figure 6.21** Smart home devices

» use of **chatbots** that interact through instant messaging, artificially replicating patterns of human interactions using AI to respond to typed or voice messages; when a question is asked, the chatbot responds using the information known at the time:



▲ **Figure 6.22** Chatbots

» autonomous cars (see Section 6.2)

» facial expression recognition
  – algorithms identify key facial landmarks such as the corners of the eyebrows, corners of the mouth, and so on
  – a combination of these landmarks can be used to map emotions (such as anger, fear, joy and surprise).

Find out more

The Turing Test is a method to test a machine's ability to match human intelligence levels. Find out how the ***Turing Test*** is used.

6.3.3 AI systems

This section considers two types of AI system:

- expert system – a computer system that mimics the decision-making ability of a human; expert systems use AI to simulate the judgement and behaviour of a human or organisation that has expert knowledge and experience
- machine learning – this is the science of training computers with sample data so that they can go on to make predictions about new unseen data, without the need to specifically program them for the new data.

Expert systems

Expert systems are a form of AI that has been developed to mimic human knowledge and experiences. They use knowledge and inference to solve problems or answer questions that would normally require a human expert.

For example, suppose the user was investigating a series of symptoms in a patient. The expert system would ask a series of questions, and the answers would lead to its diagnosis. The expert system would explain its reasoning with a statement such as '*impaired vision, lack of coordination, weak muscles, slurred speech and the patient used to work in a paint factory – the diagnosis is mercury poisoning*' – the user could then probe deeper if necessary.

The expert system will supply a conclusion and any suggested actions to take and it will also give the percentage probability of the accuracy of its conclusions (for example, the following statement could be made '*Based on the information given to me, the probability of finding oil bearing rocks in location 123AD21G is about 21%*').

There are many applications that use expert systems:

- oil and mineral prospecting
- diagnosis of a patient's illness
- fault diagnostics in mechanical and electronic equipment
- tax and financial calculations
- strategy games, such as chess
- logistics (efficient routing of parcel deliveries)
- identification of plants, animals and chemical/biological compounds.

Expert systems have many advantages:

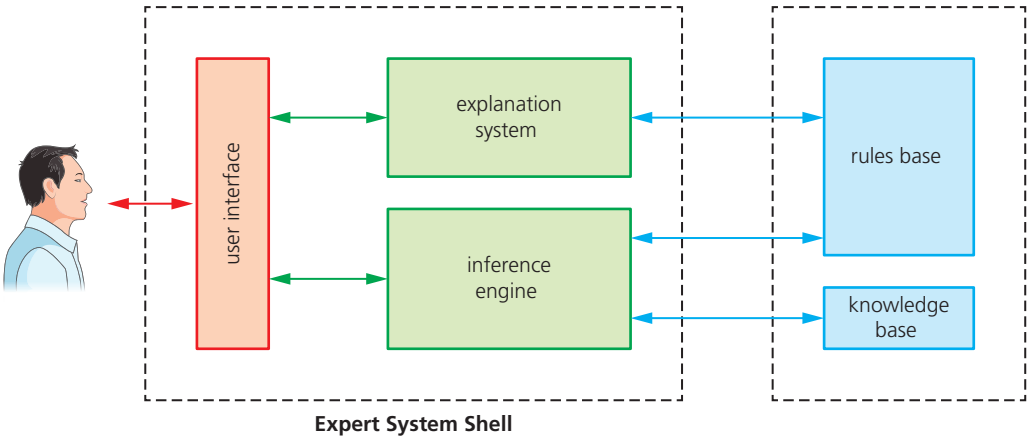
- they offer a high level of expertise
- they offer high accuracy
- the results are consistent
- they have the ability to store vast amounts of ideas and facts
- they can make traceable logical solutions and diagnostics
- it is possible for an expert system to have multiple expertise
- they have very fast response times (much quicker than a human expert)
- they provide unbiased reporting and analysis of the facts
- they indicate the probability of any suggested solution being correct.

Expert systems also have disadvantages:

- users of the expert system need considerable training in its use to ensure the system is being used correctly

- the set up and maintenance costs are very high
- they tend to give very 'cold' responses that may not be appropriate in certain medical situations
- they are only as good as the information/facts entered into the system
- users sometimes make the very dangerous assumption that they are infallible.

So what makes up an expert system? Figure 6.23 shows the typical structure of an expert system:



▲ **Figure 6.23** Expert system structure

Advice

Explanation systems are not explicitly covered in the syllabus (Figure 6.23).

Find out more

Using Figure 6.23, find out how the explanation system works and why it is included in a typical expert system.

User interface

- method by which the expert system interacts with a user
- interaction can be through dialogue boxes, command prompts or other input methods
- the questions being asked usually only have Yes/No answers and are based on the responses to previous questions.

Inference engine

- this is the main processing element of the expert system
- the **inference engine** acts like a **search engine** examining the **knowledge base** for information/data that matches the queries
- it is responsible for gathering information from the user by asking a series of questions and applying responses where necessary; each question being asked is based on the previous responses
- the inference engine is the problem-solving part of the expert system that makes use of **inference rules** in the **rules base**
- since the knowledge base is a collection of **objects** and **attributes,** the inference engine attempts to use information gathered from the user to find an object that matches (making use of the rules base to find a match)

Knowledge base

- the knowledge base is a repository of facts
- it stores all the knowledge about an area of expertise obtained from a number of expert resources

- it is basically a collection of **objects** and their **attributes**; for example:

Object	Attribute 1	Attribute 2	Attribute 3	Attribute 4	Attribute 5	Attribute 6
dog	mammal	can be a pet	lives on land	makes bark sounds	body is covered in fur	walks on 4 legs
whale	mammal	not a pet	lives in water	makes sonic sound	body covered in skin	swims; no legs
duck	bird	not a pet	lives in water	makes quack sounds	body covered in feathers	swims; has two legs

- so if we had a series of questions:
  - is it a mammal? YES
  - can it be a pet? NO
  - does it live in water? YES
  - does it make sonic sounds? YES
  - is its body covered in skin? YES
  - does it have any legs? NO

  conclusion: it is a WHALE.

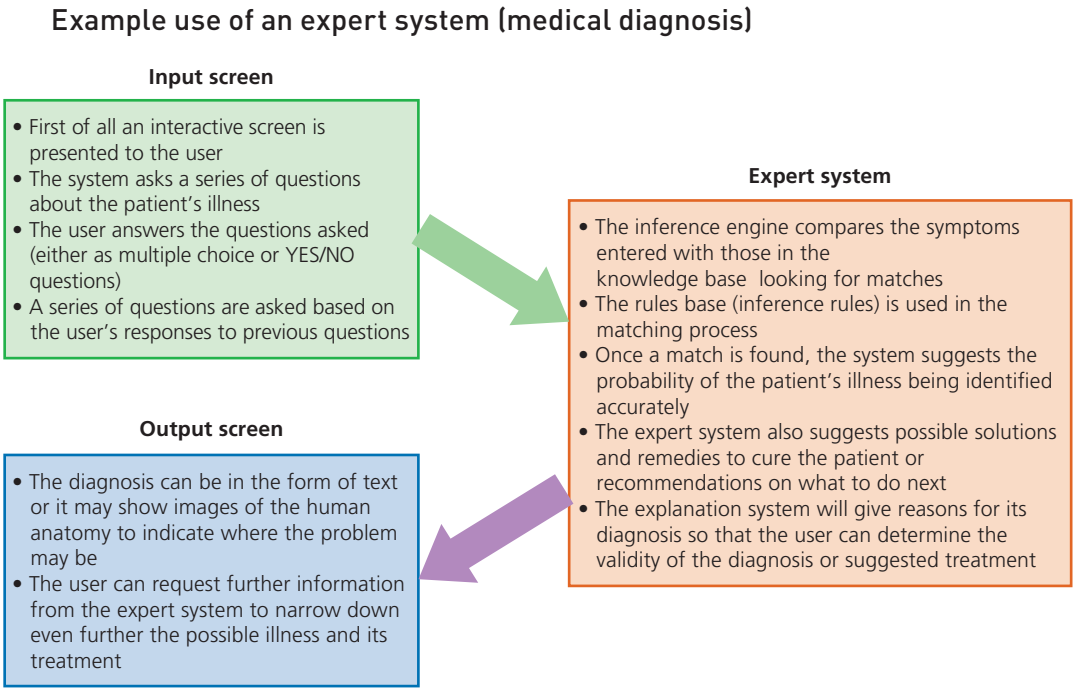
Rules base

- the rules base is a set of inference rules
- inference rules are used by the inference engine to draw conclusions (the methods used closely follow human reasoning)
- they follow logical thinking like the example above; usually involving a series of 'IF' statements, for example:

IF continent = "South America" AND language = "Portuguese" THEN country = "Brazil"

Setting up an expert system

- information needs to be gathered from human experts or from written sources such as textbooks, research papers or the internet
- information gathered is used to populate the knowledge base that needs to be first created
- a rules base needs to be created; this is made up of a series of inference rules so that the inference engine can draw conclusions
- the inference engine itself needs to be set up; it is a complex system since it is the main processing element making reasoned conclusions from data in the knowledge base
- the user interface needs to be developed to allow the user and the expert system to communicate
- once the system is set up, it needs to be fully tested; this is done by running the system with known outcomes so that results can be compared and any changes to the expert system made.



▲ **Figure 6.24** Use of an expert system

Activity 6.12

Which computer terms, connected to AI, are being described below?

a a repository of facts made up of a collection of objects and their attributes

b informs the user of the reasoning behind the expert system's conclusions and recommended actions

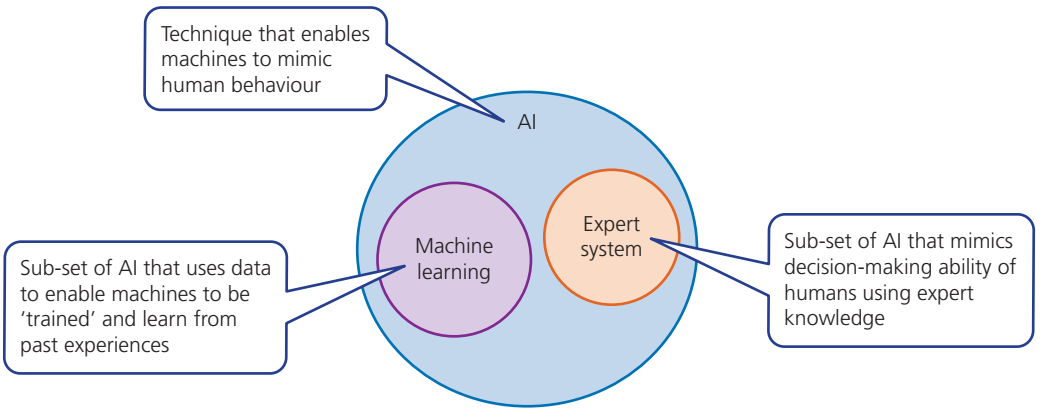
c made up of user interface and inference engine

d a sub-set of AI where machines mimic human activities and can manipulate objects using end-effectors all the way through to autonomous vehicles

e contains a set of inference rules.

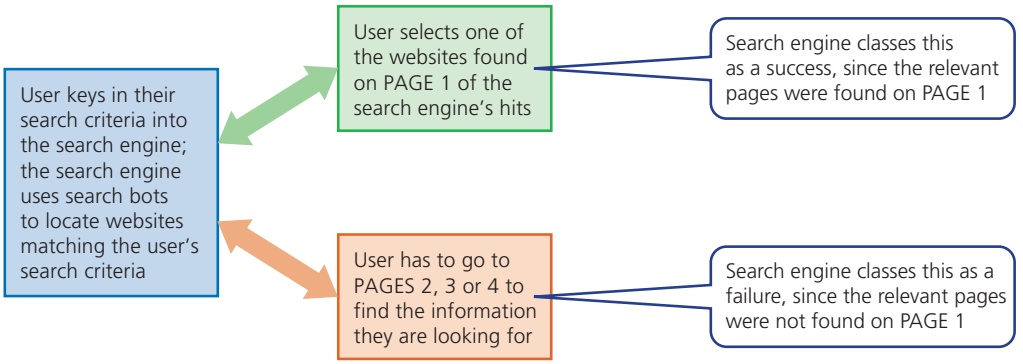
Machine learning

Recall the AI 'family' as shown in Figure 6.25.



▲ **Figure 6.25** AI family

Machine learning is a sub-set of artificial intelligence (AI), in which algorithms are 'trained' and learn from their past experiences and examples. It is possible for the system to make predictions or even take decisions based on previous scenarios. They can offer fast and accurate outcomes due to very powerful processing capability. One of the key factors is the ability to manage and analyse considerable volumes of complex data; some of the tasks would take humans years to complete without the help of machine learning techniques. One example that uses machine learning are the most sophisticated ***search engines***:



▲ **Figure 6.26** Search engine success or failure

The search engine will learn from its past performance, meaning its ability to carry out searches becomes more and more sophisticated and accurate.

Differences between AI and machine learning

▼ **Table 6.5** Difference between AI and machine learning

AI	Machine learning
represents simulated intelligence in machines	this is the practice of getting machines to make decisions without being programmed to do so
the aim is to build machines that are capable of thinking like humans	the aim is to make machines that learn through data acquisition, so that they can solve new problems

Examples of machine learning

? Example 1: Categorising email as spam

Consider email messages such as '*You have won $2 million in the National Lottery*'; how can machine learning determine that this email should be put into your spam folder?

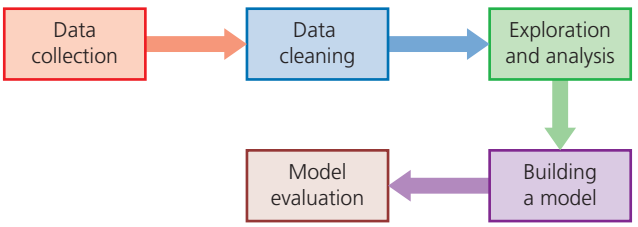
- A machine learning algorithm collects data about emails, such as email content, headers, senders name/email address and so on.
- It carries out a 'cleaning' process by removing ***stop words*** (for example, the, and, a) and punctuation, leaving only the relevant data.
- Certain words/phrases are frequently used in spam (for example, lottery, earn, full-refund) and indicate that the incoming email is very likely to be spam.
- The machine learning model is built and a 'training data set' is used to train the model and make it learn using past email known to be spam.
- Once it is evaluated, the model is fine-tuned and tested live.

Example 2: Recognising user buying history

When you visit an online retailer, such as Amazon, you might receive the message '*customers who bought Hodder Education IGCSE ICT textbook also bought Hodder Education IGCSE Computer Science textbook*'. How is machine learning used to establish a user's buying characteristics?

- This comes from ***collaboration filtering***, which is the process of comparing customers who have similar shopping behaviour to a new customer who has similar shopping behaviour.
- Suppose customer 'A' is very interested in playing football and they also bought a jazz CD, a book on Roman history and some health food.
- Two weeks later, customer 'B' who likes to go cycling also bought a similar jazz CD and a book on ancient Roman history.
- The machine learning algorithms will then recommend that customer 'B' might like to buy some health food due to the similarities between 'A' and 'B's shopping behaviour.
- This technique is particularly popular when asking your mobile phone to generate a playlist from your music library based on a few criteria you might select.

Example 3: Detection of fraudulent activity



▲ **Figure 6.27** Machine learning model development

- Data is gathered by a survey or **web scraping**; for example, to detect credit card fraud, information about customers is gathered, such as types of transactions (does the customer buy designer clothes?), shopping habits and certain personal data.
- Redundant data is then removed; this needs to be carefully done to remove the possibility of wrong predictions.
- The most important machine learning step: the algorithm is trained through real examples of customer purchasing behaviour.
- A model is built based on learning from the training data, and the machine learning algorithm can now be used to detect fraud (for example, if a customer spends an unusual amount on a piece of jewellery, there is a high chance a fraudulent activity has taken place).
- The machine learning model is then fully tested with known data and known outcomes; the system is modified if it hasn't met its criteria to detect fraudulent activity.

Activity 6.13

1 Use words/phrases from the following list to complete the paragraph below:

Word list:

artificial intelligence	expert systems	machine learning
attributes	explanation system	robotics
cognitive	inference engine	rules base
database	inference rules	search engine

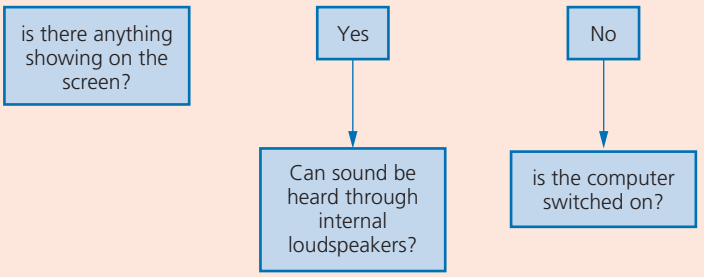
..... is a branch of computer science where the ..... function of the human brain is studied. ..... are a branch of AI which mimic the knowledge and experience of humans. The application uses an ..... to explain its reasoning and logic to the user. The main element of this application is an ..... which acts like a ..... by applying ..... to a knowledge base.

2 John has bought an expert system to help him find faults in computer systems. John has been asked to look at a computer that won't play music through some external speakers, attached to a computer using a USB port.

- The music is stored in a file on the solid state drive (SSD).
- The computer uses a sound card to output sound.
- The external loud speaker is plugged into a USB port.

a The expert system asks John a series of questions. Each question is dependent on his response to a previous question.

For example:



▲ **Figure 6.28**

Write a further series of questions that helps John identify that there is actually a fault in the external loud speaker plugged into his computer.

b Describe three other uses of expert systems.

3 a Define the term ***machine learning***.

b Explain how ***machine learning*** and ***artificial intelligence (AI)*** differ.

c Describe how a search engine might use machine learning to determine the most appropriate results based on a user's search criteria.

4 Six statements are shown on the left and nine computer terms on the right. Draw lines to link each statement with its correct computer term.

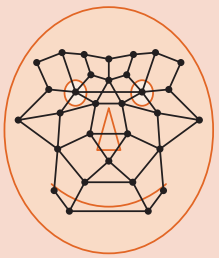
branch of computer science where cognitive behaviour of the human brain is studied	rules base
when a machine shows superior performance to a human in many tasks	search engine
application that uses knowledge and inference to solve problems that would require human expertise	strong AI
repository of facts that is a collection of objects and their attributes	artificial intelligence
contains the inference rules used to determine any matches between input data and stored data	machine learning
sub-set of AI in which the algorithms are 'trained' and learn from past experience and examples	robotics
	expert system
	knowledge base
	user interface

Extension

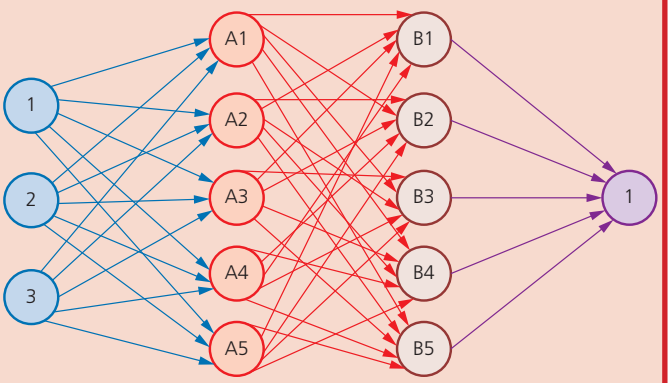
For those students considering the study of this subject at A Level, the following section gives some insight into further study on a sub-set of machine learning called deep learning

Deep learning

Deep learning structures algorithms in layers (input layer, output layer and hidden layer(s)) to create an artificial neural network made up of 'units' or 'nodes', which is essentially based on the human brain (i.e. its interconnections between neurons). Neural network systems are able to process more like a human and their performance improves when trained with more and more data. The hidden layers are where data from the input layer is processed into something that can be sent to the output layer. Artificial neural networks are excellent at tasks that computers normally find hard. For example, they can be used in face recognition:

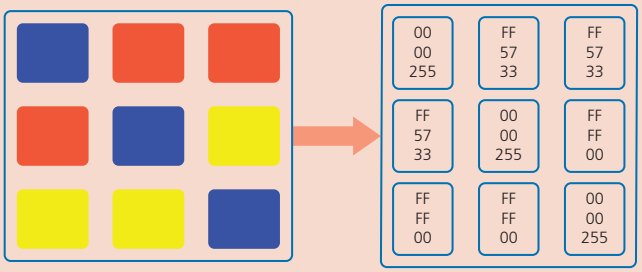


The following diagram shows an artificial neural network (with two hidden layers) – each circle, called a unit or node, is like an 'artificial neuron':



Neural networks are effective at complex visual processing such as recognising birds, for example, by their shape and colour. There are many different sizes, colours and types of bird, and machine learning algorithms struggle to successfully recognise such a wide variety of complex objects. But the hidden layers in an artificial neural network allow a deep learning algorithm to do so.

A deep learning system can perform visual processing by analysing pixel densities of camera images of objects. Consider the following small section of an image, where each pixel on the left is represented by its RGB value in hexadecimal:



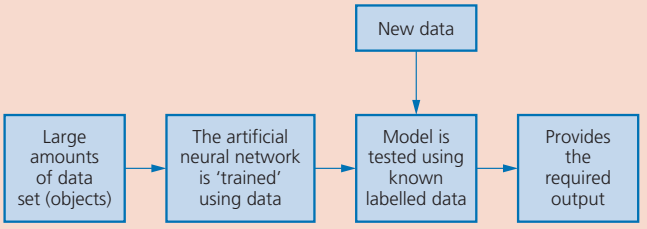
Deep learning using artificial neural networks can be used to recognise objects by looking at the binary codes of each pixel, thus building up a picture of the object. For example, the following image shows a close up of part of a face where each pixel can be assigned its binary value; patterns of binary codes can be recognised by deep learning algorithms as a person's face.



Link

For more on representing pixels in hexadecimal see Section 1.2.

Example showing how deep learning works:



Large amounts of data are input into the model. One of the methods of object recognition, using pixel densities, was described above. The nodes in the artificial neural networks learn certain parameters, based on the training data. Once this has been done, labelled data is entered (this is data which has already been defined and is therefore recognised) into the model to make sure it gives the correct responses. If the output isn't sufficiently accurate, then the model is refined until it gives satisfactory results – this might mean changing the number of nodes or number of layers. The refinement process may take several 'adjustments' until it provides reliable and consistent outputs.

Comparison between machine learning and deep learning

Machine learning	Deep learning
uses algorithms to enable machines to make decisions on their own based on past data	uses algorithms arranged as an artificial neural network to enable machines to make decisions
tends to need less data than deep learning – though this can still mean large amounts of data are required to carry out the training	the system needs large amounts of data during the training stages
is very good at certain classification tasks such as spam email	is very good at very complex non-linear tasks such as visual processing
a modular approach is taken to solve a given problem/task; each module is then combined to produce the final model	the problem is solved from beginning to end as a single entity
does not always need a large amount of computing power – however for complex problems machine learning can still require a lot of computing power	deep learning requires a large amount of computing power
testing of the system takes a long time to carry out	testing of the system takes much less time to carry out
there are clear rules which explain why each stage in the model was made	since the system makes decisions based on its own logic, the reasoning behind those decisions may be very difficult to understand (hence why they are often referred to as a ***black box***)